

Lewis Structures (Kelter) — step 0 for (1–8)

- Sum valence e- (= **group #**; add 1 per -, remove 1 per + for ions)
- Sum "wants" (table; period 3+ central: see expanded octet below)
- Bonds = (wants - valence) ÷ 2 (round down)
- Central atom = lowest EN (**never H**); ions go in brackets, charge outside
- Remaining e- = valence - 2 × bonds: outer atoms first, then central; double/triple bond if central lacks octet or nonzero FC
- FC = valence - (lone e- + # bonds): want overall 0, else - on most EN atom

	H	Be	B, Al	C, Si	N, P	O, S	F-I	Noble
Val e-	1	2	3	4	5	6	7	8
Wants	2	4	6	8	8	8	8	8

- Expanded octet** — period 3+ central (P, S, Cl, Br, I, Xe) can hold >8, so step 3 can undercount bonds. If bonds < # outer atoms: ignore step 3 — single-bond **every** outer atom, complete outer octets, then put **all leftover e- on the central** as lone pairs
- ClF₃: 28 e- - 3 bonds (6) = 9 F LP (18) = 4 left → **2 LP on Cl** → 5 groups → T-shaped. Likewise SF₄ (1 LP, seesaw); XeF₄ (2 LP, sq. planar); XeF₂, I₃- (3 LP, linear); PCl₅, SF₆ (0 LP)

Periodic Trends

	Trend	Top	Bot	Left	Right
Atomic size (radius)	✓ (Cs)	↓	↑	↑	↓
EN (electronegativity)	↗ (F)	↑	↓	↓	↑
IE (ionization energy)	↗ (He)	↑	↓	↓	↑
EA (electron affinity)	↗ (Cl)	↑	↓	↓	↑

- Cation < neutral < anion; ↑ΔEN → ↑ bond dipole (3–4); smaller ion, higher charge → ↑ΔH_{hydr} (18); larger atom → ↑ dispersion (9–12)

Trend	Bond	Enthalpy	Exception
EA (X + e- → X-)	form	exothermic	endo: Be, Mg, noble, EA2 (usually)
IE	break	endothermic	—

- Lattice energy** U ∝ Q₁Q₂ / (r₁ + r₂): heat **released** (exo.) forming ionic solid from gaseous ions (*dissociation* = endo); max U = ↑ charges, ↓ radii; used in ΔH_{solv} (18)

(1–2) Molecular Geometry (VSEPR)

- Central atom** BG + LP (double/triple bond or radical e- = 1 group); lone pairs compress angles (SCI₂ a little < 109.5°); compound examples in the (3–4) table

EG	e- geo.	Hyb.	BG	LP	Mol. geo.	Angle	Polarity
2	linear	sp	2	0	linear	180°	nonpolar
3	tri. planar	sp ²	3	0	tri. planar	120°	nonpolar
3	tri. planar	sp ²	2	1	bent	<120°	polar
4	tetra.	sp ³	4	0	tetra.	109.5°	nonpolar
4	tetra.	sp ³	3	1	tri. pyr.	<109.5°	polar
4	tetra.	sp ³	2	2	bent	<109.5°	polar
5	tri. bipy.	sp ³ d	5	0	tri. bipy.	90/120°	nonpolar
5	tri. bipy.	sp ³ d	4	1	seesaw	90/120°	polar
5	tri. bipy.	sp ³ d	3	2	T-shaped	90, 180°	polar
5	tri. bipy.	sp ³ d	2	3	linear	180°	nonpolar
6	octa.	sp ³ d ²	6	0	octa.	90°	nonpolar
6	octa.	sp ³ d ²	5	1	sq. pyr.	90°	polar
6	octa.	sp ³ d ²	4	2	sq. planar	90°	nonpolar

(3–4) Polarity & Dipole Moment

Compounds	BG	LP	Shape	Polarity
diatomic: H ₂ , Br ₂ (same); CO, IBr, HCl (diff.)	1	—	linear	nonpol. / pol.
BeCl ₂ , CO ₂ , CS ₂	2	0	linear	nonpolar
BF ₃ , BH ₃ , SO ₃	3	0	tri. planar	nonpolar
SO ₂ , O ₃ , ClNO	2	1	bent	polar
CH ₄ , CCl ₄ , CF ₄ , SiH ₄ , PO ₄ ³⁻ , hydrocarb.	4	0	tetra.	nonpolar
CH ₃ Cl/F/Br/I, CH ₂ Cl ₂ , CHCl ₃ (mixed)	4	0	tetra.	polar
NH ₃ , PH ₃ , AsH ₃ , PCl ₃ , NCl ₃ , NF ₃	3	1	tri. pyr.	polar
H ₂ O, OF ₂ , H ₂ S, SCI ₂ , NH ₂ -	2	2	bent	polar
PCl ₅ , PBr ₅	5	0	tri. bipy.	nonpolar
ClF ₃	3	2	T-shaped	polar
XeF ₂ , I ₃ -	2	3	linear	nonpolar
BrF ₅	5	1	sq. pyr.	polar
XeF ₄	4	2	sq. planar	nonpolar

- Dipole** = separation of + / - charge
- (μ ≠ 0) = **polar** = bond dipoles (ΔEN) **don't cancel** (lone pairs or mixed outer atoms)
- CH₃F > CH₃Cl > CH₃Br > CH₃I; NH₃ > PH₃ > AsH₃ (EN)
- Polar dissolves polar: NH₃ in H₂O ✓, H₂S in NH₃ ✓; CCl₄, SiO₂, O₂, CO₂ ✗
- Solvent polarity: H₂O > ethanol > acetone > CH₂Cl₂ > diethyl ether > toluene > hexane
- Alkanes C_nH_{2n+2}, all **nonpolar** — C count: meth 1, eth 2, prop 3, but 4, pent 5, hex 6, hept 7, oct 8, non 9, dec 10

(5–6) Hybridization of an Atom in a Structure

- Look up the EG **of that atom** in the (1–2) table: 2 → sp, 3 → sp², etc.
- C, 4 single bonds → **sp³**; C with 1 double bond (C=O, C=C, benzene ring) → **sp²**; C with triple or 2 doubles → **sp**
- O in -O-H or ether (2 atoms + 2 LP) → **sp³**; carbonyl =O → **sp²**
- N with 3 atoms + 1 LP (amine) → **sp³**
- Each atom separately: CH₃-CHO → left C sp³, carbonyl C sp²

(7–8) σ and π Bonds: Counting & Concepts

- single** = 1σ; **double** = 1σ + 1π; **triple** = 1σ + 2π
- σ = hybrid orbitals, head-on along the bond axis; π = unhybridized p-p, above & below the axis
- cis** = same side, **trans** = opposite sides; **π restricts rotation** (double/triple bonds)
- CH₃-CH=CH-COOH: 11σ, 2π; benzene C₆H₆: 12σ (6C-C + 6C-H) + 3π
- Two 1s (or head-on p+p along bond axis) → σ and σ*; sideways p+p (perpendicular axis) → π and π*
- Benzene delocalized π MOs: e- **free to move around the ring**

(9–10) Intermolecular Forces

Force	Found in
Dispersion (London, van der Waals, induced dipole)	ALL molecules; ↑ w/ size & molar mass; straight-chain > branched
Dipole-dipole	polar molecules
H-bond	H bonded directly to N, O, F
Ion-dipole	ion + polar solvent (NaCl in H ₂ O)

Compounds	Disp.	Dip.-dip.	H-bond
Br ₂ , Cl ₂ , F ₂ , H ₂ , CO ₂ , BCl ₃ , CF ₄ , hydrocarb., noble gases	✓	✗	✗
HCl, HBr, SO ₂ , CHCl ₃ , CH ₃ Br, CH ₃ CHO, CH ₃ CN, ethers (CH ₃ OCH ₃), H ₂ S, CH ₃ SCH ₃	✓	✓	✗
H ₂ O, NH ₃ , HF, CH ₃ OH, CH ₃ CH ₂ OH, CH ₃ NH ₂ , CH ₃ COOH, H ₂ O ₂ , HOCH ₂ CH ₂ OH	✓	✓	✓

(11–12) Rank Vapor Pressure & Boiling Point

- Ionic > H-bond > dipole-dipole > dispersion** (at similar size)

IMF	bp	mp	Visc.	Surf. tens.	ΔH _{vap}	Crit. T	VP
↑	↑	↑	↑	↑	↑	↑	↓
↓	↓	↓	↓	↓	↓	↓	↑

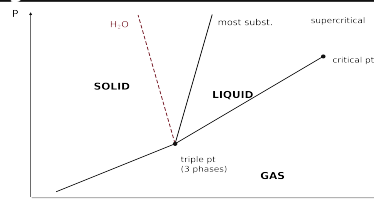
Rank	Example
VP ↓	CF ₄ > PF ₅ > BrF ₃ (polar = lowest VP)
bp ↑	CH ₄ < CH ₃ Cl < CH ₃ OH < RbCl (ionic)
bp (disp. size)	I ₂ > Br ₂ > Cl ₂ > F ₂ ; CH ₃ CHO > CH ₃ OCH ₃ > Rn
bp (H-bond)	HF > HI > HBr > HCl
bp (# of OH)	HOCH ₂ CH ₂ OH > ClCH ₂ CH ₂ OH > CH ₃ CH ₂ OH > CH ₃ CH ₂ Cl

(13) Viscosity, Capillary Action, Surface Tension

Surface tension	energy required to stretch/increase surface area by unit amount
Viscosity	resistance of a liquid to flow ; ↑ with IMF, ↓ with ↑T; spherical/compact → lower
Capillary action	liquid flows up a thin tube against gravity = cohesion (like molecules) + adhesion (liquid-glass)

- Concave meniscus (H₂O): adhesion > cohesion; convex (Hg): cohesion > adhesion. Highest viscosity = most H-bonding (HOCH₂CH₂OH)

(14) Phase Diagrams



- Lines = 2 phases coexist; **triple point** = all 3; **critical point**: above T_c no pressure can liquefy gas → **supercritical fluid**
- ↑T at const P (→): solid → liquid → gas. ↑P at const T (↑): gas → liquid → solid. Normal bp/mp read at 1 atm

Endothermic (+ΔH, +ΔS)	Exothermic (-ΔH, -ΔS)
melting s→l	freezing l→s
vaporization l→g	condensation g→l
sublimation s→g	deposition g→s

- Order: solid = greatest, gas = least (↑T = ↓ order)
- H₂O**: fusion line leans **left** (ice less dense) → ↑P melts ice. **CO₂**: triple pt at 5.1 atm → at 1 atm solid **sublimes** s→g (dry ice)

(15–16) Type of Solid

Type	Held by	Properties	Examples
Ionic	± attraction	hard, brittle, high mp, poor cond.	NaCl, AgCl, CaF ₂ , MgO, CaCO ₃
Molecular	IMFs	soft, low mp, poor cond.	ice H ₂ O, CO ₂ , I ₂ , sucrose
Metallic (atomic)	e- sea	low-high mp, good cond.	Au, Cu, Fe, Na
Network cov. (atomic)	covalent bonds	v. hard, v. high mp, poor cond.	diamond, quartz SiO ₂ , graphite, SiC
Nonbonding (atomic)	dispersion	v. low mp	noble gases: Ar(s)
Amorphous	no long-range order	melts over a range	glass, rubber, plastic, butter, wax
Ceramic	inorganic nonmetal, fired	brittle, heat-resistant	clay/kaolinite, porcelain, Al ₂ O ₃

- Graphite sp² sheets: conducts along the sheets, slippery; diamond sp³: insulator
- FCC packing = most space-efficient (12 neighbors)

(17) Band Theory & Doping

- Bands = MOs delocalized over the whole crystal: **valence band = bonding MOs, conduction band = antibonding MOs, band gap** between
- Conductor** (metal): bands continuous/overlap; **Semiconductor** (metalloids Si, Ge): small gap, e- promotable; **Insulator: LARGE gap**
- Doping** = add impurity to ↑ conductivity:

	Dopant	Carrier
n-type	Group V/5A (P, As)	extra e- in conduction band (– charges)
p-type	Group III/3A (B, Al, Ga, In)	holes in valence band (e- hop between holes)
p-n junction	—	diode : current flows one way

(18) Heat of Solution & Hydration

$$\Delta H_{\text{solution}} = \Delta H_{\text{hydration}} - \Delta H_{\text{lattice energy}}$$
$$\Delta H_{\text{solution}} = \Delta H_{\text{solute}} + (\Delta H_{\text{solvent}} + \Delta H_{\text{mix}})$$
$$= (-\Delta H_{\text{lattice}}) + \Delta H_{\text{hydration}}$$

- Dissolving = break crystal (**ΔHsolute = –ΔHlattice**, costs energy) + make ion-dipole attractions (**ΔHsolvent + ΔHmix = ΔHhydration**, releases)
- ΔHhydration**: heat **released** when 1 mol **gaseous ions** dissolves in water (always exo)
- Highest ΔHhydr**: smallest ion + highest charge: Al³⁺ > Mg²⁺ > Na⁺ > K⁺; F[–] > Cl[–]
- ΔHsoln **exo (–)**: ΔHhydration > ΔHsolute; **endo (+)**: solute > hydration; **≈ 0**: comparable

(19) Saturation; Like Dissolves Like

Unsaturated	less than equilibrium amount — more can dissolve
Saturated	max amount at that T; dissolved ⇌ solid dynamic equilibrium
Supersaturated	more than equilibrium amount; unstable — seed crystal/scratch → excess crystallizes out

- Solution forms if solute-solvent attraction **comparable** to solute-solute + solvent-solvent; polar/ionic ⇌ polar (H₂O, ethanol, NH₃); nonpolar ⇌ nonpolar (hexane, CCl₄, benzene)
- Most soluble in ethanol = most H-bonds (ethylene glycol); in hexane = longest chain (1-pentanol); all **nitrates** soluble

(20) Solubility vs P & T; When Bubbles/Boiling

↑ gives	Gas	Solid / liquid
↑ Pressure	↑ solubility (Henry)	no effect
↑ Temperature	↓ solubility	↑ solubility

- Dissolve the most gas: **low T + high P** (cold soda holds CO₂)
- Liquid **boils** when its vapor pressure = **external (atm) pressure**; gas **bubbles out** of solution when its partial (vapor) pressure > atm pressure

(21) Molarity (and Molality)

$$M = \frac{\text{mol solute}}{L \text{ solution}} \quad m = \frac{\text{mol solute}}{\text{kg solvent}}$$

- mL ÷ 1000 = L; g ⇌ mol via molar mass; dilution: M₁V₁ = M₂V₂
- mol = M × V(L): 0.500 M × 0.250 L = **0.125 mol**
- V = mol ÷ M: 0.300 mol ÷ 0.450 M = 0.667 L = **667 mL**
- Molality m uses **kg of solvent**, not solution — for ΔT_f/ΔT_b
- Only molarity changes with temperature** (volume swells); molality, mass %, χ don't (mass/count-based)

(22) Percent Concentration (% w/v)

$$\% w/v = \frac{g \text{ solute}}{100 \text{ mL soln}} \Rightarrow g = \frac{\% \times mL}{100}$$

- g = % × mL ÷ 100: for 250 mL of 1.5% → 1.5 × 250 ÷ 100 = **3.75 g** (the summary's "31.25 g" contradicts its own formula)
- mL = g ÷ (% ÷ 100): 16.6 g ÷ 0.0450 g/mL = **369 mL**
- % m/m** = g solute ÷ (g solute + g solvent) × 100 — denominator = **whole solution**
- g water = g solution × (1 – % ÷ 100): 300 g of 9.71% KBr → 300 × 0.9029 = **271 g water**
- % v/v** = vol component ÷ total vol × 100

(23) Henry's Law (C_{gas} = k_H·P_{gas})

$$C_{\text{gas}} = k_H P_{\text{gas}} \quad k_H = C_{\text{gas}} / P_{\text{gas}}$$

- k_H = "solubility constant of a gas" (M/atm); C = molar solubility
- k_H = C ÷ P** (divide — below 1 atm, k_H comes out **larger** than C)
- 1 atm** = 760 mmHg = 760 torr = 14.7 psi = 101.3 kPa (mmHg → atm: ÷ 760)

Given C = 4.7e–4 mol/L at 522 mmHg:

- P = 522 ÷ 760 = 0.687 atm
- k_H = 4.7e–4 ÷ 0.687 = **6.8e–4 mol/L·atm**

(24) Osmosis

- Osmosis: solvent** (water) flows across a semipermeable membrane (passes solvent, **blocks solute** ions — it does *not* stop water) from **low solute conc.** → **high solute conc.**
- Hyperosmotic** = more solute: water flows **into** it (cell shrinks)
- Hyposmotic** = less solute: water flows **out** to the more concentrated side (cell swells)
- Isosmotic** = equal: no net flow (isotonic saline = 0.9% w/v NaCl); n = pressure needed to **stop** osmotic flow

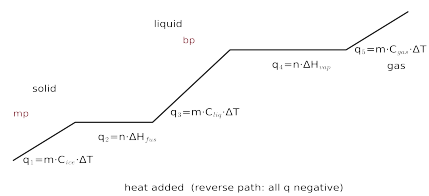
(25) MO Theory — Bond Order & Magnetism

	B2, C2, N2 pattern	O2, F2, Ne2 pattern
high	σ*2p	σ*2p
↑	n*2p, n*2p	n*2p, n*2p
↑	σ2p	n2p, n2p
↑	n2p, n2p	σ2p
↑	σ*2s	σ*2s
low	σ2s	σ2s
Used by	B2, C2, N2, N2+, N2–	O2, O2+, O2–, F2, Ne2; NO, NO+, NO– (heteronuclear: use the more EN atom's pattern); Cl2, Cl2+, Cl2– (period 3: n = 3)

- 1s 2s 2p 3s 3p 4s 3d 4p 5s 4d 5p 6s 4f 5d 6p 7s 5f 6d 7p
- 1. Valence e- total: **add** 1 per – charge, **remove** 1 per +
- 2. Fill round-robin sublevel before double occupation (Hund)
- 3. **BO = (bonding e- – antibonding e-) / 2**
- 4. Any unpaired e- → **paramagnetic**; all paired → **diamagnetic**
- 5. BO > 0 → stable, can form; **higher BO = stronger & shorter bond** (N₂ > N₂⁺). Counting 2p alone gives same BO (σ_{2s} + σ*_{2s} cancel)
- 6. **Heteronuclear** (NO, CO, OF–): sum both atoms' valence e-, fill the **O2 pattern**. NO 11 e- → BO 2.5, para; NO+ 10 → 3, dia; NO- 12 → 2, para
- 7. Two-sided diagram: **more EN atom's orbitals sit lower** (more stable); bonding MOs resemble it

Sp.	e-	BO	Mag	Sp.	e-	BO	Mag
N ₂	10	3	dia	O ₂	12	2	para (2↑)
N ₂ ⁺	9	2.5	para	O ₂ ⁺	11	2.5	para
N ₂ [–]	11	2.5	para	O ₂ [–]	13	1.5	para
F ₂ , Cl ₂	14	1	dia	Cl ₂ ⁺	13	1.5	para
B ₂	6	1	para	Cl ₂ [–]	15	0.5	para

(26) Phase-Change Heat (gas ⇌ solid)



- Sketch the path; skip segments you don't cross (see examples below)
- Sloped (one phase): **q = m(g)·Cs·ΔT**; flat (phase change): **q = n(mol)·ΔH**
- Sum; convert J → kJ. Cooling path (gas → solid): every q **negative** (ΔHcond = –ΔHvap, ΔHfreeze = –ΔHfus)
- N₂: C(ice) 2.09, C(liq) 4.184, C(steam) ≈ 2.0 J/g·°C; ΔHfus 6.02, ΔHvap 40.7 kJ/mol; MM 18.02

36.0 g H₂O, liquid 65 °C → gas 115 °C:

- heat liquid 65→100: q = 36.0 g × 4.184 × 35° = 5.27 kJ
- boil (use mol): q = (36.0 ÷ 18.02) × 40.7 = 81.3 kJ
- heat gas 100→115: q = 36.0 g × 2.01 × 15° = 1.09 kJ
- total = **87.7 kJ**
- Cooling runs the same steps backward: 125 g benzene (MM 78.11), gas 425 K → liquid 335 K (bp 353, mp 279 – it never freezes): 9.5 kJ + 54.3 kJ + 3.9 kJ = **67.7 kJ removed**
- ΔHsub = ΔHfus + ΔHvap (endo/exo table in (14))

(27) Raoult's Law (VP Lowering)

$$P_{\text{soln}} = \chi_{\text{solvent}} P_{\text{solvent}}^{\circ} \quad \Delta P = \chi_{\text{solute}} P_{\text{solvent}}^{\circ}$$
$$\chi_{\text{solvent}} = \frac{\text{mol}_{\text{solvent}}}{\text{mol}_{\text{solvent}} + i \cdot \text{mol}_{\text{solute}}}$$

- mol solvent = g ÷ MM; mol solute = g ÷ MM (**× i if ionic**)
- χ(solvent) = mol solvent ÷ total mol — always < 1
- P(soln) = χ(solvent) × P°; lowering ΔP = P° – P(soln) = χ(solute)·P°
- 75.0 g citric acid (192.12) + 420 g H₂O, P° 71.93 mmHg: 0.390 & 23.31 mol → χ = 0.9835 → **70.7 mmHg**

(28) Freezing ↓ / Boiling ↑ (van't Hoff)

$$\Delta T_f = i K_f m \quad \Delta T_b = i K_b m$$

- m = mol solute / **kg solvent**; H₂O: K_f 1.86, K_b 0.512 °C/m; T_f = T°_f – ΔT_f, T_b = T°_b + ΔT_b
 - i** = particles per formula unit: glucose, sucrose 1; NaCl, KCl, NaOH 2; CaCl₂, (NH₄)₂SO₄ 3; Al(NO₃)₃ 4. Given the bp: i = ΔT_b ÷ (K_b × m)
 - Lowest fp for equal grams = small MM and high i (NaOH), **but the HW answer key rewards the most ions per formula (CaCl₂, i = 3)**
 - Molar mass from fp — 20.0 g unknown per 500 g benzene (T°_f 5.444, K_f 5.12) freezes at 3.77 °C:
- ΔT_f = 5.444 – 3.77 = 1.674 °C
 - m = ΔT_f ÷ (i × K_f) = 1.674 ÷ 5.12 = 0.327 mol/kg
 - mol = m × kg solvent = 0.327 × 0.500 = 0.163 mol
 - MM = g ÷ mol = 20.0 ÷ 0.163 = **≈120 g/mol**

(29) Osmotic Pressure → Molar Mass

$$\pi = i M R T \Rightarrow M = \frac{\pi}{i R T}$$

- R = **0.08206** L·atm/mol·K; T in **K** (+273.15); n in **atm** (torr ÷ 760); i = 1 nonelectrolyte

Given 15.2 **mg** in 150.0 mL, n = 8.44 torr, 25 °C:

- M = n ÷ (iRT) = (8.44 ÷ 760) ÷ (0.08206 × 298) = 4.54e–4 M
- mol = M × V(L) = 4.54e–4 × 0.1500 = 6.81e–5 mol
- MM = g ÷ mol = 0.0152 ÷ 6.81e–5 = **223 g/mol**